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Molecular dynamics simulations of aqueous ionic clusters using polarizable water

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The solvation properties of a chlorine ion in small water clusters are investigated using state-of-the-art statistical mechanics. The simulations employ the polarizable water model developed recently by Dang [J. Chem. Phys. **97**, 2659 (1992)]. The ion–water interaction potentials are defined such that the successive binding energies for the ionic clusters, and the solvation enthalpy, bulk vertical binding energy, and structural properties of the aqueous solution agree with the best available results obtained from experiments. Simulated vertical electron binding energies of the ionic clusters $Cl^-(H_2O)_n$, $(n=1–6)$ are found to be in modest agreement with data from recent photoelectron spectroscopy experiments. Minimum energy configurations for the clusters as a function of ion polarizability are compared with the recent quantum chemical calculations of Combariza, Kestner, and Jortner [Chem. Phys. Lett. **203**, 423 (1993)]. Equilibrium cluster configurations at 200 K are described in terms of surface and interior solvation states for the ion, and are found to be dependent on the magnitude of the Cl^- polarizability assumed in the simulations.

I. INTRODUCTION

An accurate description of ion–water interactions remains a central problem in the chemical physics of solvation.[1,2] A number of computational studies have been undertaken to examine ionic systems, with most focusing only on the properties of ionic solutions.[3–7] An alternative approach, considered here, is to investigate ion–water clusters. Extensive information on the structural and energetic properties of small ion–water clusters, with typically six or fewer water molecules, is available from both experiment[8,9] and from accurate *ab initio* calculations.[1,10] This information may be used in the development of simple and computationally efficient interaction models, which in turn may be applied to studies of large clusters and bulk solution. An analysis of solvation properties as a function of cluster size should then reveal further details of the nature of ionic solvation in aqueous solutions.

Earlier, we reported a study in which classical molecular dynamics computer simulations were used to examine the structural and energetic properties of the ionic clusters $Cl^-(H_2O)_n$, $(n=1–4)$.[11] An empirical, polarizable model was used to represent water, while ion–water interactions were defined by fitting to experimental binding enthalpy data.[9] From finite temperature simulations, it was observed that the chlorine ion sits typically on the surface of the water clusters. Perera and Berkowitz carried out molecular dynamics simulations for larger clusters using the same potential model.[12] They found that the chlorine ion is only partially solvated even with water molecules numbering up to 20. This observation is in conflict with ideas that ion solvation in clusters may be described in terms of a sequential filling of solvation shells.[13,14] Recently, Combariza, Kestner, and Jortner[1] reported *ab initio* quantum chemical calculations of the halogen anion–water clusters $X^-(H_2O)_n$, (X=Cl,Br,I, and $n=1–6$). In particular, they investigated the relative stability of possible surface and interior configurations for the anions. They concluded that, while surface states are favored for the smaller clusters, interior states become energetically competitive at around $n=6$. These results appear to be in conflict with the observations of Perera and Berkowitz for larger Cl^-–water clusters.

As part of our research on water and ionic solutions, we believe it is important to understand the origin of this conflict. Progress toward this goal is reported in the present paper. Minimum energy configurations are calculated for $Cl^-(H_2O)_n$, $(n=1–6)$ using the molecular dynamics potential model. This allows for a direct comparison with the *ab initio* calculations of Combariza, Kestner, and Jortner. We have also investigated the role of ion polarizability in the molecular dynamics simulations. Finally, the relationship between the calculated minimum-energy configurations and ion solvation properties at finite temperature is discussed for $Cl^-(H_2O)_6$. The paper is organized as follows. In Sec. II, the potential and methodology are described. The results and discussion are presented in Sec. III, and the conclusions are given in Sec. IV.

II. POTENTIAL MODEL AND SIMULATIONS

The interaction energy in our molecular dynamics model consists of the Lennard-Jones and electrostatic interaction between water–water and water–ion pairs and a nonadditive polarization energy term. The total interaction energy is written as

$$U_{\text{tot}}=U_{\text{pair}}+U_{\text{pol}}, \tag{1}$$

where the pairwise additive potential is given as a sum of interactions between pairs of centers i and j,

$$U_{\text{pair}}=\sum_i \sum_j \left(\frac{A_{ij}}{r_{ij}^{12}}-\frac{C_{ij}}{r_{ij}^{6}}+\frac{q_iq_j}{r_{ij}}\right) \tag{2}$$

with each pair potential expressed as a sum of a Lennard-Jones interaction with parameters A_{ij} and C_{ij} and a Cou-

TABLE I. Potential parameters for models of water, Cl^- ion in water: Lennard-Jones parameters (σ and ϵ), charges q, and atomic polarizabilities α.[a]

Atom type	σ (Å)	ϵ (kcal/mol)	q	α (Å^3)
O	3.205	0.160	−0.7300	0.528
H			+0.3650	0.170
Cl^-	4.455	0.100	−1.0000	3.690[b]
Cl	3.816	0.100	0.0000	2.180[c]

[a]The water–water parameters are taken from Refs. 15 and 19.
[b]Reference 23.
[c]Reference 25.

lombic interaction between fixed charges q_i and q_j on the centers i and j. The polarization potential is given by

$$U_{\text{pol}}=-\frac{1}{2}\sum_i \mu_i E_i^0, \tag{3}$$

where μ_i is the induced dipole on center i and E_i^0 is the electrostatic field at center i arising from all other fixed charges in the system. The induced dipole moment μ_i and the total electrostatic field E_i at the polarizable center j are evaluated self-consistently from the following expressions:

$$\mu_i=\alpha_i E_i \tag{4}$$

and

$$E_i=E_i^0+\sum_{j\neq i} T_{ij}\mu_j, \tag{5}$$

where α_i is the polarizability of atom i and T_{ij} is the dipole tensor.

The procedure used for computing the photoelectron spectra for the ionic cluster has been discussed in detail elsewhere.[15] The pertinent equation is stated below,

$$E_{\text{stab}}(n)=BE_v(n)-\text{EA}=-\langle (U_{\text{tot}}^i-U_{\text{tot}}^n)\rangle_{\text{Cl}^-(\text{H}_2\text{O})_n}. \tag{6}$$

$E_{\text{stab}}(n)$ and $BE_v(n)$ are the stabilization and photodetachment energies, respectively, and EA is the electron affinity. Thus, the stabilization energy can be obtained from a single simulation, where at each configuration the difference between the total potential for the ionic and neutral clusters is computed. We have successfully used the procedure outlined above in a recent study of photoelectron spectra for the solvated iodine anion.[15]

We constructed the parameters for the ion–water and neutral chlorine–water interaction potentials so that the computed enthalpies of binding of the clusters $Cl^-(H_2O)_n$, (n=1–4), agree with the experimental results of Kebarle *et al.*,[9] and the computed solvation enthalpy, the bulk vertical binding energy,[16] and structural properties of the ionic solution reproduce experimental results.[17] The final parameters are listed in Table I. For all ion–water cluster simulations, the dynamics were performed in the microcanonical ensemble, and the rotational and translational motions were removed only at the beginning of the simulation. For each complex, simulations are performed with 20 ps of equilibration followed by 500 ps of data collection. For the ionic solution simulation, the sample consisted of one single ion and 215 water molecules in a cubic cell of 18.6 Å. We used temperature and pressure coupling constants of 0.2 and 0.5,[18] respectively, and a time step of 1.5 fs. The SHAKE (Ref. 18) procedure was adapted to constrain all the bond lengths to their equilibrium values, and the nonbonded interactions were cut off at 9 Å. The simulation consisted of 75 ps of equilibration followed by 150 ps of data collection for averaging.

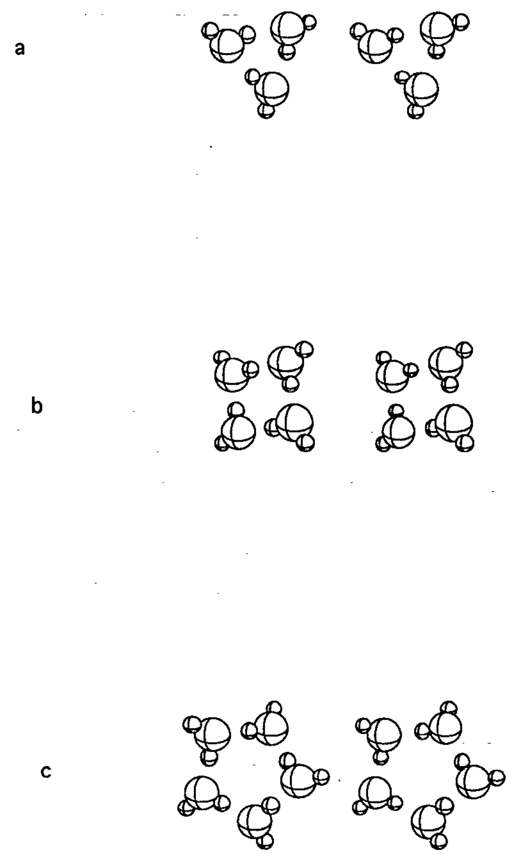

FIG. 1. Minimum energy configuration of water clusters (a) trimer; (b) tetramer; and (c) pentamer.

III. RESULTS AND DISCUSSION

A. Water clusters

Before discussing ionic cluster simulations, it is important to demonstrate the effectiveness of the polarizable water model of Dang[19] in representing water clusters. This is accomplished through comparison with accurate *ab initio* quantum chemical calculations of small water cluster properties. Minimum energy configurations for water clusters up to the pentamer have been determined through quenching of multiple configurations obtained from simulation trajectories at an average temperature of 200 K. The trimer, tetramer, and pentamer structures are displayed in Fig. 1. All have a cyclic geometry, with each water molecule acting as both a hydrogen bond donor and acceptor. The O–O bond lengths for these clusters are given in Table II. The calculated geometries of the trimer and tetramer

TABLE II. Comparison of properties of water clusters.[a]

Water cluster	Refs. 15 and 19 E_b	Refs. 15 and 19 R_{oo}	Ref. 21 E_b	Ref. 21 R_{oo}	Ref. 20 E_b	Ref. 20 R_{oo}
Trimer	−15.42	2.79	−15.20	2.78	−13.87	2.80
Tetramer	−27.25	2.71	−27.90	2.72	−24.33	2.74
Pentamer	−37.07	2.70				

[a]The minimum binding energies, E_b are in kcal/mol and R_{oo} is the minimum oxygen–oxygen distance, in Å, between the nearest-neighbors.

are in excellent agreement with those predicted from both the accurate *ab initio* calculations of Xantheas and Dunning, Jr.,[20] and from the sophisticated pair-potential model of Reimers and Watts,[21] as shown in Table II. The cluster binding energies are also displayed in the table, and are found to be in reasonable agreement with, although somewhat overestimating the energies predicted by *ab initio* calculations.[20]

B. Thermodynamic and structural properties of ionic clusters and ionic solutions

We begin our discussion of ionic clusters by presenting Table III, in which the calculated binding energies of the clusters $Cl^-(H_2O)_n$, ($n=1$–4) are compared with the experimental results of Kebarle *et al.*[9] As can be seen from the table, the agreement between calculated and experimental results is quite good. The minimum energy structures for the clusters, displayed in Fig. 2, are similar to those reported in previous molecular dynamics simulations.[11,12] Note that the present model does not include explicit three-body water–ion–water interactions. Nevertheless, the same level of accuracy as reported in previous works[11,12] has been obtained. Using these nonadditive potentials, the selective structural and thermodynamic properties for a single Cl^- ion in aqueous solution have been calculated and then compared with results from other computer simulations and the available experimental data for these systems. The ion–water radial distribution functions, calculated using the nonadditive potentials, are shown in Fig. 3. The peak positions of g_{Cl-O} and g_{Cl-H} are similar to the results of computer simulations reported in the literature.[3–6] The peak heights are shorter, however, and the coordination number around the chlorine ion is reduced. This results directly from the ion polarizability.[22]

TABLE III. Calculated and experimental enthalpies of binding (eV) of $Cl^-(H_2O)_n$ at 300 K.

n	MD	Expt.[a]
1	−0.58±0.02	−0.57
2	−1.11±0.04	−1.11
3	−1.61±0.06	−1.62
4	−2.10±0.07	−2.10
5	−2.51±0.08	···
6	−2.89±0.08	···

[a]Reference 9.

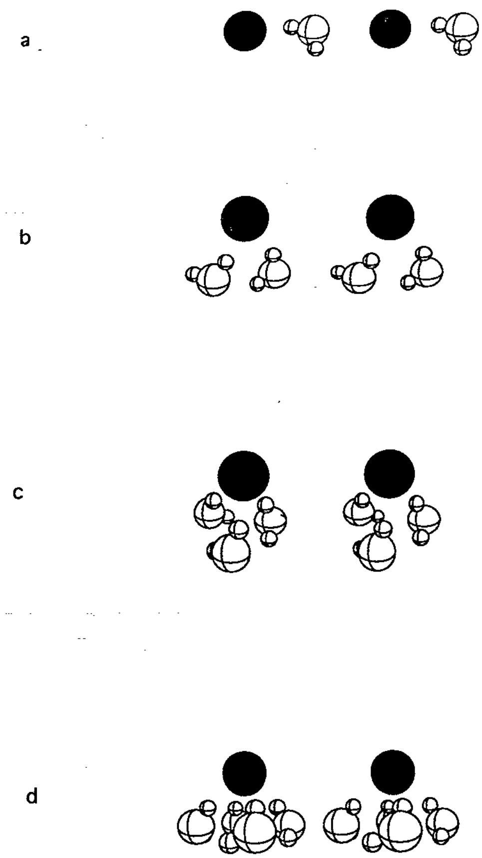

FIG. 2. Minimized structures of ionic clusters (a) $Cl^-(H_2O)$; (b) $Cl^-(H_2O)_2$; (c) $Cl^-(H_2O)_3$; and (d) $Cl^-(H_2O)_4$.

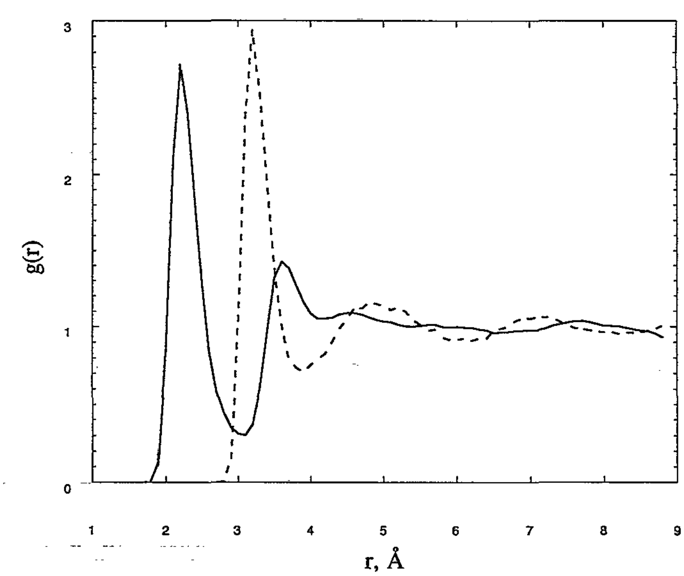

FIG. 3. Calculated Cl–O radial distribution function g_{Cl-O} (dashed line) and Cl–H radial distribution function g_{Cl-H} (solid line) obtained from molecular dynamics simulations of an aqueous solution of Cl^- at room temperature.

TABLE IV. Structural and thermodynamic properties of the Cl^- ion in water at 300 K.

	MD	Expt.
R_{ClO} (Å)	3.20	3.20–3.34[a]
R_{ClH} (Å)	2.20	2.22–2.26[a]
Coordination number	6.8	5.3–6.0[a]
ΔH_{sol} (eV)	−3.41	−3.42[b]
I_v (eV)[d]	9.30	8.90[c]

[a]Reference 26.
[b]Reference 17.
[c]Reference 16.
[d]Bulk vertical photodetachment energy obtained from molecular dynamics simulations using Eq. (6).

During the molecular dynamics simulations, a neutral chlorine–water potential was constructed using Eq. (6) so that the simulated bulk vertical binding energy reproduced experimental measurements.[16] A summary of the calculated solution-phase properties is compiled in Table IV, along with results from experimental work on comparable systems. It seems clear that the present model reliably represents the thermodynamic and structural properties of ionic clusters and ionic solutions.

C. Photoelectron spectra of ionic clusters

Chlorine ion–water cluster stabilization energies as determined from Eq. (6) are presented in Table V along with the experimental results of Markovich *et al.*[8] The molecular dynamics results are presented at two temperatures, 0 and 200 K, as there is considerable uncertainty in the temperature of the experimental measurements. The 0 K simulation results are obtained by calculating the stabilization energy for minimum energy structures determined for the various clusters. The agreement between the simulation and experimental data presented in Table V is poor for $n=1$ and 2, but reasonable for larger clusters. There is some disagreement between different experimental measurements on the $Cl^-(H_2O)$ cluster, with Kebarle *et al.*[9] and Markovich *et al.*[8] reporting binding energies of −0.57 and −0.71 eV, respectively. Our ion–water potential model is based on the energetic properties reported by Kebarle *et al.* Furthermore, the simulation results are also in accordance with those reported from quantum chemical calculations by Combariza, Kestner, and Jortner.[1] It is interesting that for the solvated I^- ion, calculated vertical binding energies for all clusters sizes were in good agreement with the corresponding experimental data.[15]

TABLE V. Stabilization energies, $E_{stab}(n)$, for $Cl^-(H_2O)_n$ clusters obtained from molecular dynamics simulations [Eq. (6)] and experiment.

		MD (eV)	
n	Experiment (eV)[a]	0 K[b]	200 K
1	0.71	0.58	0.56
2	1.32	1.13	1.09
3	1.84	1.70	1.59
4	2.24	2.20	1.98
5	2.54	2.52	2.32
6	2.86	2.87	2.55

[a]Reference 8.
[b]Minimum energy configurations.

TABLE VI. Averaged potential energies (eV) and their components for the $Cl^-(H_2O)_n$.

n	$-U_{pair}$	$-U_{pol}$	$-U_{tot}$
1	0.42	0.14	0.56
2	0.83	0.24	1.06
3	1.20	0.33	1.53
4	1.55	0.44	1.99

D. Comparison with *ab initio* data: The role of the Cl^- polarizabilities

As mentioned above, one of the major goals of this study is to compare the structures obtained from the molecular dynamics model with the *ab initio* calculations reported by Combariza, Kestner, and Jortner.[1] In previous works,[11,12] and in the work described in this paper, we find that structures in which the anion is located on the surface of the water cluster are most favorable. This finding seems to be inconsistent with the recently published results of Combariza, Kestner, and Jortner. The *ab initio* studies suggest the existence of surface and interior isomers of comparable energy for the ionic cluster $Cl^-(H_2O)_6$. The surface configuration they describe consists of a hexagonal ring of water molecules with the anion located above the water plane. The interior configuration is a distorted octahedron consisting of two cyclic water trimers with the anion sandwiched symmetrically between the water groups.

Simulated annealing and multiple quenching techniques have been used to identify over 40 distinct local energy minima in the $Cl^-(H_2O)_6$ system. The vast majority and the most stable of these structures correspond to surface type configurations. Some of these resemble the hexagonal pyramidal state described by Combariza, Kestner, and Jortner, while most contain one or more water molecules bonded exclusively to other waters, i.e., "second solvation shell" water molecules. Attempts to explicitly generate the interior state described above showed it to be unstable. This instability results from the significant chlorine ion polarizability, as will be discussed below.

Some insight into the origin of the observed molecular dynamics structures may be gained from an analysis of the role ion polarizability plays in these systems. In constructing the polarization ion–water potential, the chlorine polarizability has been set to 3.69 Å^3,[23] a value close to the experimentally derived polarizability for the ion in aqueous solution at infinite dilution.[24] Table VI contains a summary of the contributions of the various potential energies to the total energy of the system for the clusters $Cl^-(H_2O)_n$, ($n=1$–4). The polarization energy contributes between 20%–25% to the total potential energy of the system. To the best of our knowledge, there has been no comparable *ab initio* based decomposition of the total energy for anion water clusters into two-body and many-body contribu-

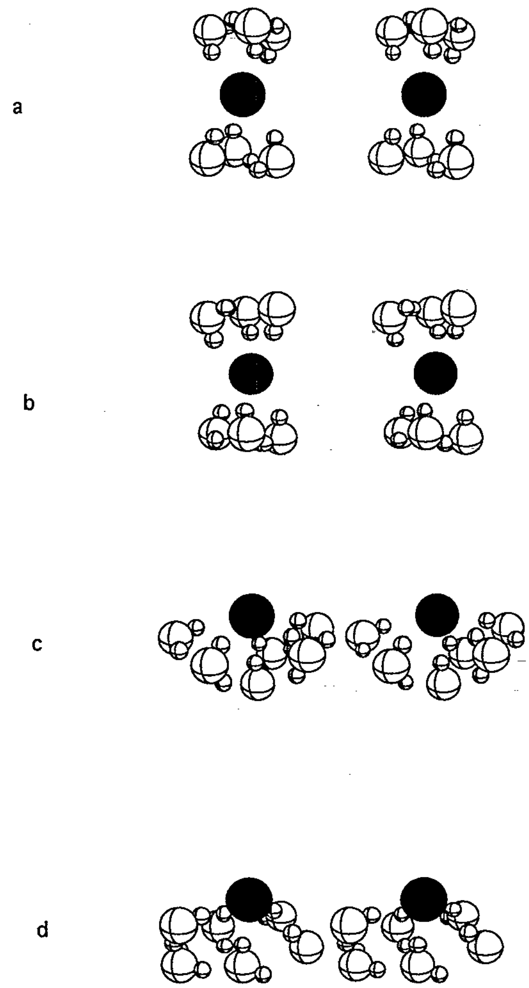

FIG. 4. Minimized structures of the ionic cluster $Cl^-(H_2O)_6$ as a function of the Cl^- polarizability. (a) $\alpha_{ion}=0$ Å^3; (b) $\alpha_{ion}=1.5$ Å^3; (c) $\alpha_{ion}=3.0$ Å^3; (d) $\alpha_{ion}=3.69$ Å^3.

tions. Such a decomposition would aid in evaluating the effectiveness of the present molecular dynamics model.

Figures 4 and 5 display locally minimized structures and energies for a series of $Cl^-(H_2O)_6$ clusters in which the ion polarizability has been varied from 0 to 3.69 Å^3. The initial configuration with zero ion polarizability is constructed to match the distorted octahedral structure of Cambariza, Kestner, and Jortner.[1] The distorted octahedron is stable for ion polarizabilities $\lesssim 2.0$ Å^3. As the ion polarizability is increased toward its experimentally derived value, the symmetric structure relaxes to a surface-type configuration. These results may be explained by noting that the water molecules experience a nearly uniform field around the ion when ion polarizability is low, and symmetric structures about the ion are therefore acceptable. A similar observation has been made for cations with small polarizabilities such as Na^+ and Li^+.[11] As the ion polarizability increases, however, the ability to form a large dipole on the ion leads to a breaking of the solvation symmetry and the formation of surface-type structures. Thus we see that the molecular dynamics potentials can reproduce the interior structure identified by Cambariza, Kestner, and Jortner, but only with a substantial reduction in the ion polarizability from its experimentally derived value. This conflict may be symptomatic of a failure of our classical electrostatic plus point-polarizability model for ionic solutions. On the other hand, the computational expense of *ab initio* calculations on systems of this size and complexity is substantial, and demonstrating the accuracy of a particular calculation is difficult. We believe that further work is necessary to fully understand these issues.

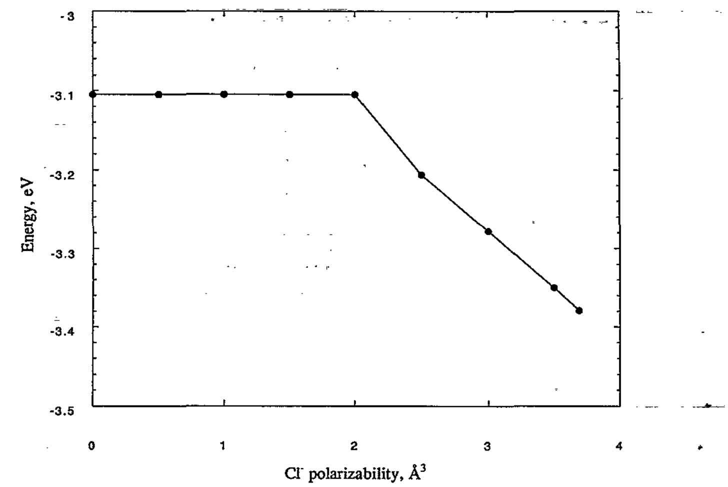

FIG. 5. Total energy of the ionic cluster $Cl^-(H_2O)_6$ as a function of the Cl^- polarizability.

E. Finite temperature cluster structures

Our primary interest in pursuing studies of ion–water clusters is to understand ionic solvation properties at finite temperatures. The relationship between minimum energy cluster configurations and average structures at finite temperatures becomes increasingly complex as cluster sizes are increased. For the $Cl^-(H_2O)_6$ cluster, as mentioned above, we have identified over 40 distinct local energy minima on the potential energy surface. In evaluating whether surface or interior configurations are more important at finite temperature, it is essential to consider not only the relative energies of "typical" states, but also their relative densities. For small clusters, the number of surface states is far greater than the number of interior states, and entropy will favor the formation of surface configurations at higher temperatures.

We have investigated structural features of the $Cl^-(H_2O)_6$ cluster at 200 K as a function of ion polarizability. In order to bracket a range of possible behaviors, α_{ion} values of 0 and 3.69 Å^3 are considered. For the zero polarizability case, the ion–water Lennard-Jones parameters have been adjusted such that the experimental binding enthalpy data of Kebarle *et al.*[9] is reproduced for the clusters $Cl^-(H_2O)_n$, ($n=1$–4). Figures 6 and 7 show results for the $\alpha_{ion}=0$ Å^3 case. Snapshots from a 1 ns trajectory of the $Cl^-(H_2O)_6$ cluster are displayed in Fig. 6, while the distance of the Cl^- ion from the cluster center of mass during the same trajectory is presented in Fig. 7. It is ev-

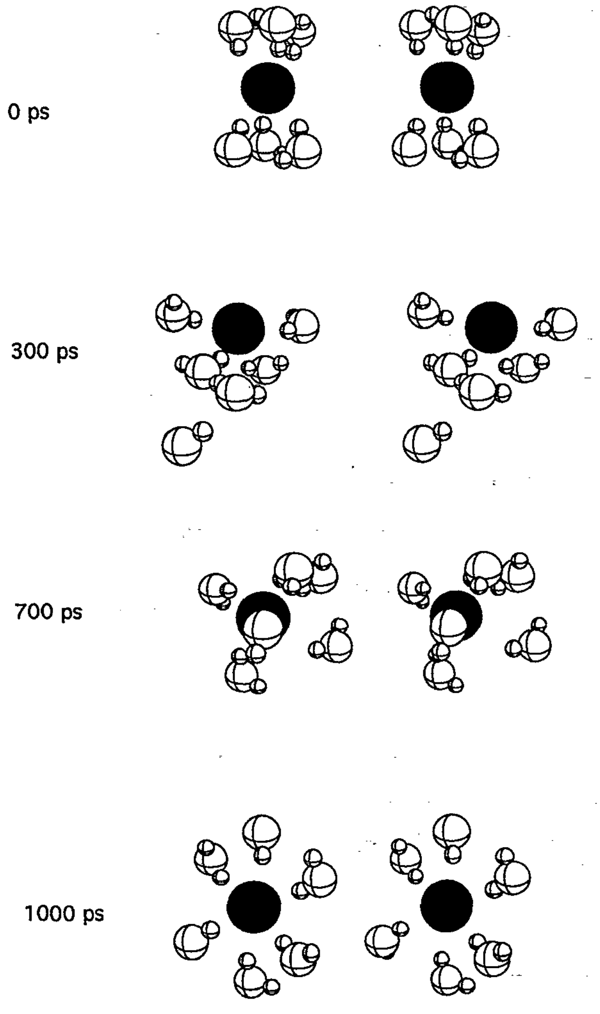

FIG. 6. Snapshots in stereo of the cluster $Cl^-(H_2O)_6$ with $\alpha_{ion}=0$ Å^3 near 200 K.

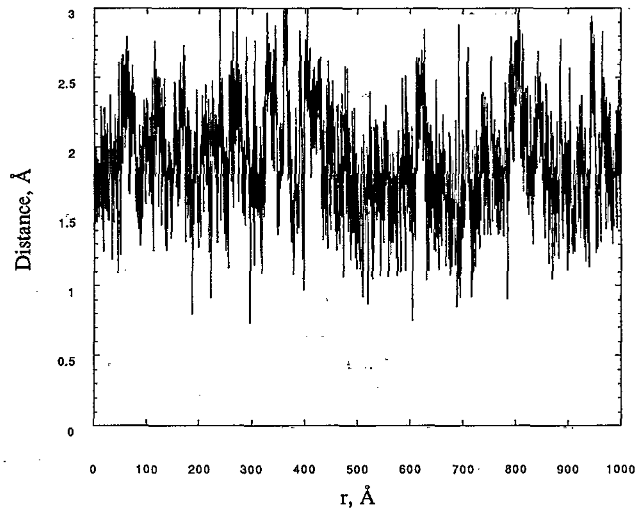

FIG. 8. Same as Fig. 7 with $\alpha_{ion}=3.69$ Å^3.

ident from both figures that there is a competition between surface and interior states. In contrast, the surface states clearly dominate for the $\alpha_{ion}=3.69$ Å^3 case, as can be seen in Fig. 8. Anion–oxygen pair correlation functions calculated from the two simulations are compared directly in Fig. 9. Note that there is substantial population in the second solvation shell in both systems. The average number of water molecules in the first solvation shell is 4.9 for $\alpha_{ion}=0$ Å^3 and 4.0 for $\alpha_{ion}=3.69$ Å^3. In summary, ion polarizability provides a significant energetic driving force for asymmetric solvation of a chlorine ion in small water clusters. Eliminating this driving force, however, does not lead to complete interior solvation of the chlorine ion at finite temperatures. This is at least in part due to entropic effects as discussed above.

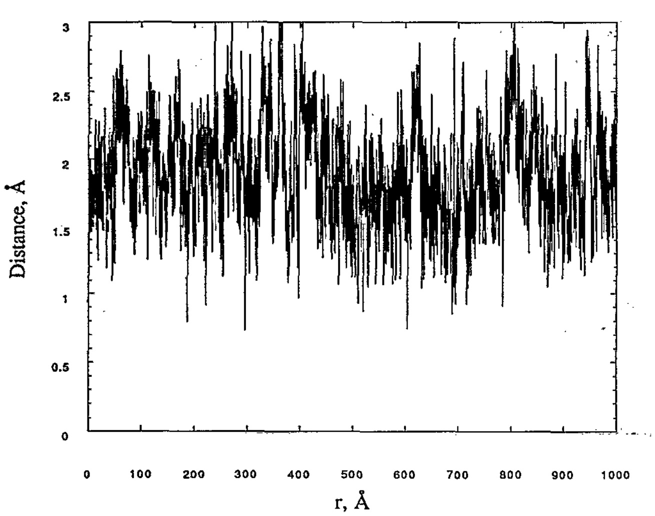

FIG. 7. The time dependence of the distance from the chlorine anion to the cluster center of mass for the cluster $Cl^-(H_2O)_6$ with $\alpha_{ion}=0$ Å^3 near 200 K.

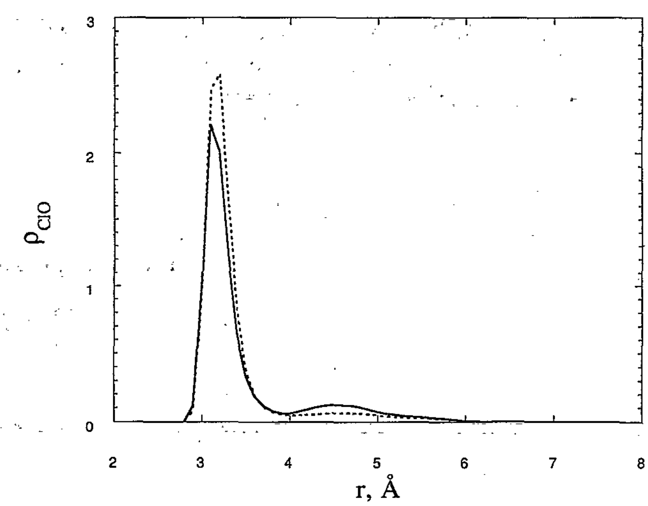

FIG. 9. Calculated atomic density $\rho_{ClO}(r)$ for the cluster $Cl^-(H_2O)_6$ with $\alpha_{ion}=0$ Å^3 (dashed line) and $\alpha_{ion}=3.69$ Å^3 (solid line) near 200 K.

IV. CONCLUSIONS

We have presented the results of a detailed computer simulation study of the solvation properties of a chlorine ion in polarizable water. The classical, polarizable potential model used in the study has been constructed such that the thermodynamic and structural properties of the ionic clusters and ionic solutions reproduce accurately the best available experimental data. The model used in this study provides a level of accuracy comparable to that reported in previous studies, without including three-body, water–ion–water interactions.

Stabilization energies for the clusters $Cl^-(H_2O)_n$, ($n=1-6$) have been computed and are found to be in modest agreement with values obtained from photoelectron spectroscopy experiments. Minimum energy configurations for the clusters have been compared with the *ab initio* structures reported recently by Combariza, Kestner, and Jortner.[1] The molecular dynamics potential model is unable to reproduce their interior, distorted octahedral structure without a substantial reduction in the polarizability of the chlorine ion. Finite temperature calculations suggest that ion polarizability is only partially responsible for the predominance of surface configurations observed in the molecular dynamics simulations. Even with zero ion polarizability, the first solvation shell around the chlorine ion contains on average less than five water molecules for the $Cl^-(H_2O)_6$ cluster at 200 K.

Further work is required to identify the origin of the discrepancy between the *ab initio* and empirical calculations. It is possible that these differences indicate some failure of our electrostatic plus point-polarizability model to accurately represent aqueous clusters. On the other hand, the quality of *ab initio* calculations on such large systems is difficult to demonstrate and questions therefore remain as to the accuracy of the calculations of Combariza, Kestner, and Jortner. Regardless of the answers to these questions, it seems unlikely at present that an accurate quantum chemical approach can deal thoroughly with the configurational complexity of clusters containing more than a few water molecules. As such, we believe that a combination of approaches that include *ab initio*, empirical or semiempirical, and experimental methods is most likely to reveal the true behavior of these fascinating aqueous systems.

ACKNOWLEDGMENT

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